

Navigating the AI, ML, DL, & LLM Landscape

From “AI” as a buzzword to a practical tool-selection framework.

TODAY'S OUTCOMES

- Define the boundary between AI, ML, DL and LLMs.
- Match common business tasks to the right paradigm.
- Explain why transformer scale created an LLM inflection point.

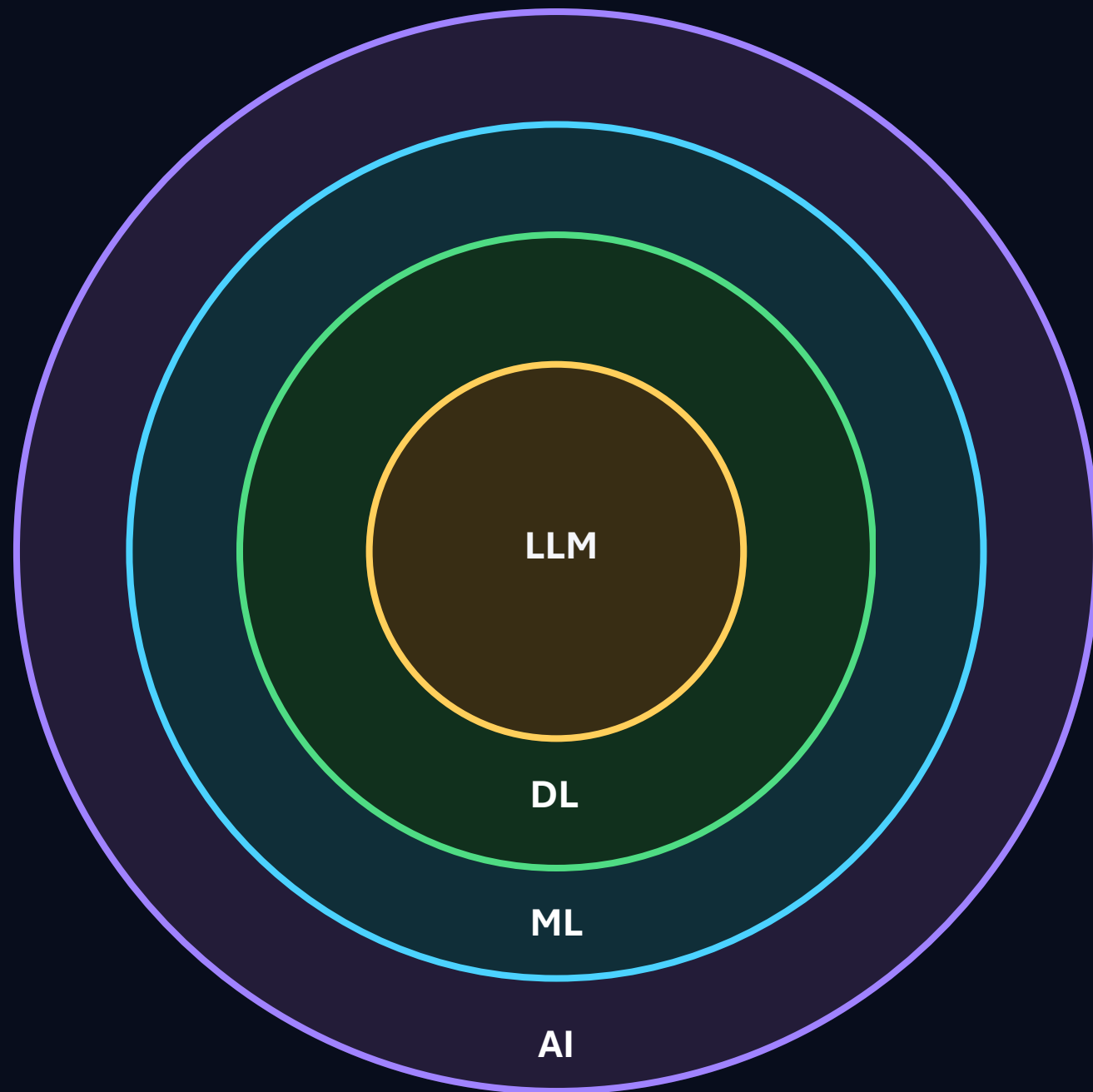
MENTAL MODEL

**Start with the
problem — not
the buzzword.**

Data shape → output → feedback signal →
appropriate tool

The Matryoshka Doll Model

Four nested paradigms — each adds a new capability layer.



AI

Simulating human intelligence

Deep Blue • Expert systems
Can include fixed rules / logic.

ML

Learning patterns from data

Linear Regression • XGBoost • Random Forest
Statistical learning often starts with engineered features.

DL

Multi-layer neural networks

CNNs • ResNet
Learns useful representations directly from complex / unstructured inputs.

LLM

Foundation neural network over massive text corpora

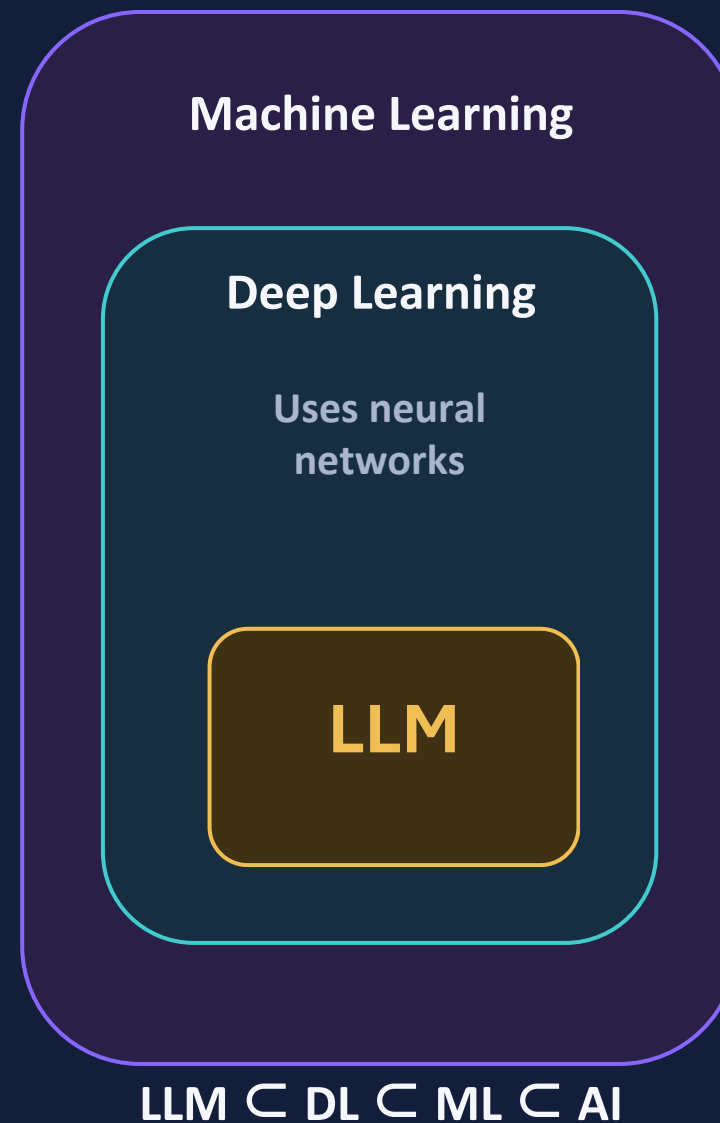
GPT-4 • Gemini • Claude • Llama 3
Generalizes through token prediction; supports zero / few-shot use.

Rule of thumb: all LLMs are DL → all DL is ML → all ML is AI. The inverse is false.

Zoom in: Neural Networks → Deep Learning → LLMs

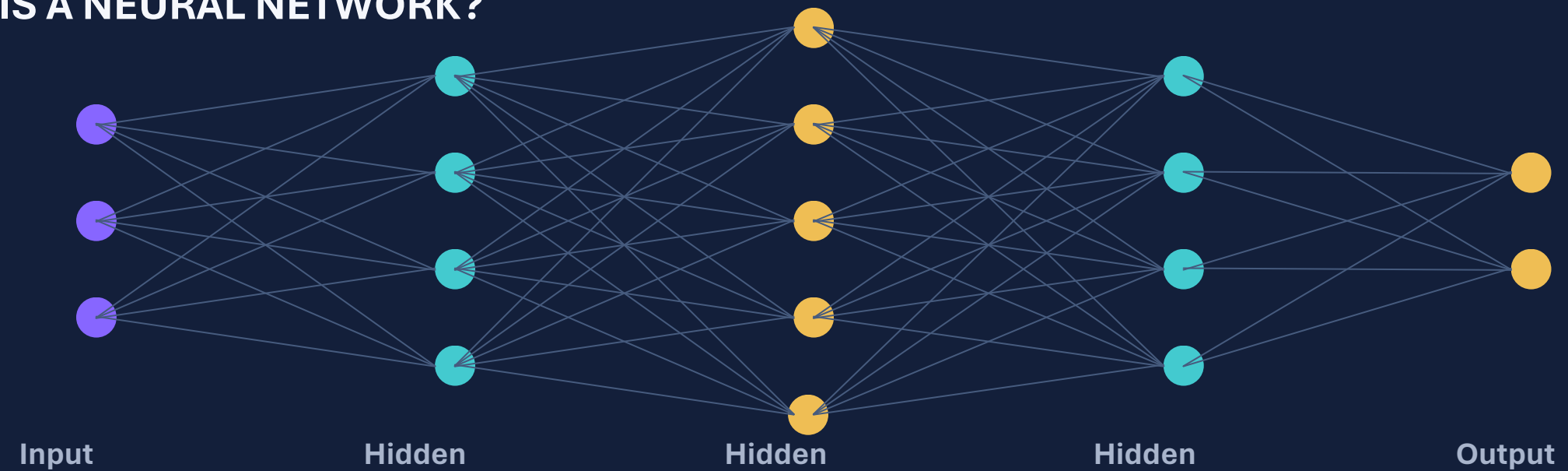
An LLM is not a different universe from neural networks — it is a very large, specialized deep neural network.

THE NESTING RULE



Think of each node as a tiny mathematical unit.

WHAT IS A NEURAL NETWORK?



1. Node

Takes inputs × weights then applies an activation.

2. Layer

Many nodes work in parallel to learn representations.

3. Depth

More stacked layers = deeper transformation of the signal.

LLM = a deep neural network (typically Transformer-based) trained at huge scale to model language by predicting tokens.

What changes as you move deeper?

The boundary is easier to remember when you compare the job each paradigm performs.

AI

“Can a machine behave intelligently?”

- Rules can be enough.
- No learning is required.

ML

“Can the pattern be learned from examples?”

- Data drives the model.
- Structured data is common.

DL

“Can the model learn its own representations?”

- Neural networks learn features.
- Images / audio / complex signals fit well.

LLM

“Can language be generated or reasoned over?”

- Massive pretraining.
- Strong zero / few-shot behavior.

Practical question to ask first

“What does the system need to learn, and what kind of data gives it that learning signal?”

Three learning modes

Start with the feedback signal: labels, no labels, or rewards.

SUPERVISED

Labeled data

The model sees examples with known answers and learns a mapping from input → target.

Examples

- Spam vs. not spam
- House price next month

Classification + Regression

UNSUPERVISED

Unlabeled data

There is no target answer. The model searches for structure, such as customer groups or lower-dimensional representations.

Examples

- Group customer behavior
- PCA / UMAP visualization

Clustering + Dimensionality reduction

REINFORCEMENT

Reward-driven

An agent acts inside an environment, observes feedback, and gradually learns which actions lead to better outcomes.

Examples

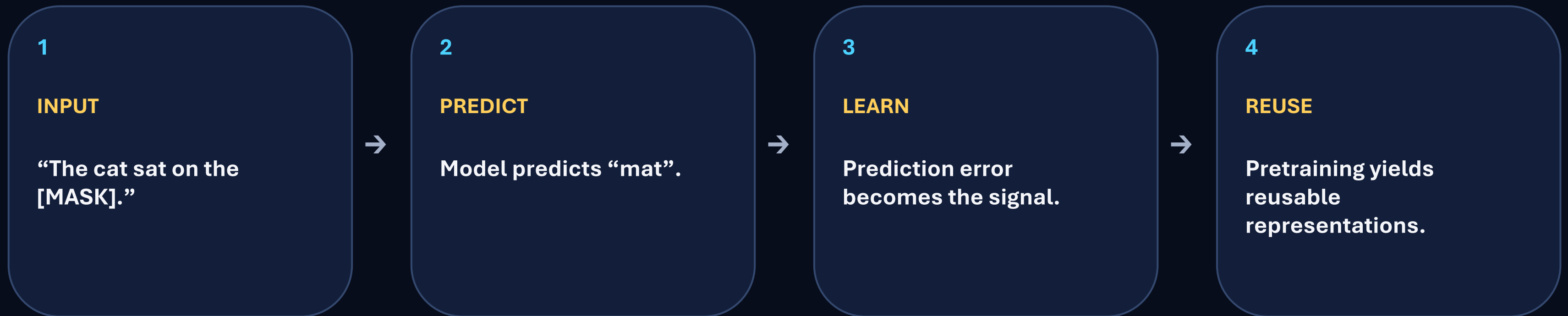
- AlphaGo
- Autonomous driving

Agent + environment + feedback

Decision cue: ask “What feedback does the model have available?”

Self-supervised learning: labels created by the data

The training signal comes from the data itself — not manual labels.



Why this changed the game

Instead of manually annotating every example, the model can learn from huge unlabeled corpora by turning prediction into its own training target.

Transformer + scale = an inflection point

Architecture unlocked scale; scale unlocked new capability regimes.



Scale is not simply “more data” — it combines compute, parameters and training data.

Three ways to adapt an LLM

Choose the lightest mechanism that gives you the behavior and knowledge you need.

METHOD	HOW IT WORKS	EFFORT	KNOWLEDGE	BEST FOR
Prompt engineering	Instruction + context window	LOW	Static	Instant tasks • general content
RAG	Retrieve external facts → pass into prompt	MEDIUM	Dynamic	Enterprise search • internal document QA
Fine-tuning	Update model weights on custom data	HIGH	Static	Tone • specialized domain syntax

Practical lens

Prompt = behavior • RAG = fresh knowledge • Fine-tuning = specialized behavior / syntax

Worked example: company policy assistant

A concrete case makes the difference between RAG and fine-tuning easier to see.

Business need: “Answer questions about our internal policy PDF and use the latest approved wording.”

PROMPT ONLY

Ask the model with instructions.
Useful when the knowledge is already in context or the task is generic.

RAG

Retrieve the relevant policy text first,
then give that evidence to the model
for grounded Q&A.

FINE-TUNE

Use custom training data to change
behavior, tone or specialized syntax
— but the knowledge itself remains
static until retrained.

For policy Q&A, the source content needs to be retrieved at question time → RAG is the natural fit.

Match the business problem to the right tool

Use the problem shape, not the buzzword, to choose the technology.

01	Predict loan default Tabular + explainability	CLASSIC ML
02	Personalized sales email Generative language	LLM
03	Detect visual defects Computer vision	DEEP LEARNING
04	Segment unlabelled customers Clustering	CLASSIC ML
05	Recommend next product Interaction history	CLASSIC ML / DL
06	Transcribe service calls Speech-to-text	DEEP LEARNING
07	Forecast electricity demand Structured time series	CLASSIC ML
08	Insider trading from chats Semantic context	LLM / NLP DL
09	Policy PDF Q&A Retrieve + synthesize	LLM + RAG
10	Find churn drivers SHAP / feature importance	CLASSIC ML + XAI

10-second method: input type → output type → feedback signal.

A practical architectural flowchart

Start with the data shape, then ask what the system must do with it.



The framework is a filter, not a rulebook: real systems can combine ML, DL, NLP and LLM components.

Walk-through: policy PDF Q&A

See how the framework produces an architecture step by step.

INPUT →	PDF policy documents
DATA TYPE →	Text / language
TASK →	Answer a natural-language question
KNOWLEDGE →	Needs exact internal source text
ARCHITECTURE	RAG → retrieve → prompt LLM → answer

Notice the reasoning path: the technology appears only after the problem and data are identified.

Enterprise takeaways

Use this as the mental model to carry into real projects.

1. Match the tool to the task

Avoid over-engineering. A simple tabular problem does not automatically need a large LLM.

2. Tabular still belongs to classic ML

Gradient-boosted trees can be fast, practical and interpretable for structured numbers.

3. LLMs shine on unstructured language

Use an LLM for generation / synthesis; add RAG when proprietary knowledge must be retrieved.

Final question for the room

“What is the simplest technology that can solve the problem well?”

Q&A • Discussion • Apply the framework to one project from your own team

Discussion

Your task: Identify the most appropriate AI/ML landscape.

Consider: What is the input data? What is the desired output? Does the system need prediction, classification, generation, retrieval, or discovery?

Question 1 — Customer Churn

A telecommunications company wants to identify customers who are most likely to cancel their subscription in the next 30 days. The company has several years of historical customer records containing age, subscription plan, monthly bill, number of complaints, usage frequency, contract duration, payment history, and whether the customer eventually cancelled. The business wants a prediction for each active customer and would also like to understand which factors contribute most to the prediction.

Question: Which AI/ML landscape would you choose, and why?

Question 2 — Quality Inspection

A manufacturing company operates a conveyor belt that produces thousands of components every day. Currently, human inspectors examine photographs of each component to determine whether it contains cracks, scratches, missing parts, or other defects. The company has collected a large image dataset containing examples of defective and non-defective components and wants to automate the inspection process in real time.

Question: Which AI/ML landscape would you choose, and why?

Question 3 — Internal Policy Assistant

A large organization has thousands of internal policies, procedures, HR documents, manuals, and guidelines stored as PDF files. Employees frequently ask questions such as, “What is the company's travel reimbursement limit?” or “What documents are required for overseas business travel?” The answers must come from the latest internal documents, and the company does not want the AI to rely only on information learned during model training.

Question: Which AI/ML landscape would you choose, and why?

Question 4 — Customer Segmentation

A retail company wants to better understand its customers but does not have predefined customer categories. It has information such as purchase frequency, average spending, product preferences, visit frequency, and transaction history. Management wants the system to automatically discover groups of customers with similar behavior so that marketing teams can design different strategies for each group.

Question: Which AI/ML landscape would you choose, and why?

Question 5 — Sales Assistant

A software company wants to help its sales team communicate with potential customers. The system receives information about the customer, such as their industry, company size, products they are interested in, previous interactions, and the salesperson's notes. It should generate a personalized follow-up email for each customer in a professional and natural writing style. The salesperson should be able to review and edit the generated email before sending it.

Question: Which AI/ML landscape would you choose, and why?